Supporting information

New Radical Salts of Bis(ethylenediseleno)tetrathiafulvalene (BEST) with Paramagnetic Tris(oxalato)metalate Anions

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Table 1S. Intra and interstack distances shorter than 3.60 Å (S···S), 3.70 Å (Se···S) or 3.80 Å (Se···Se) (the sum of the van der Waals radii), for the salt $BEST_4[Cr(C_2O_4)_3].C_6H_5COOH\ H_2O$. (**1**).

ET molecules	Interstack	Intrastack
B–B	Se(5)–Se(7)^{iv} 3.6752(11) (× 2)	
C–C	Se(10)–Se(12)^{vi} 3.6172(10) (× 2)	
A–C	Se(1)–Se(11)^{i} 3.6149(10) (× 2) Se(3)–Se(9)^{i} 3.7149(11) (× 2)	
A–D	Se(2)–Se(13)^{ii} 3.5050(11) (× 2) Se(4)–Se(15)^{ii} 3.5279(11) (× 2) Se(3)–S(13)^{i} 3.589(2) (× 2) Se(4)–S(15)^{ii} 3.512(2) (× 2) S(2)–Se(13)^{ii} 3.4725(19) (× 2) S(4)–Se(14)^{iii} 3.529(2) (× 2)	
B–C	S(8)–Se(10)^{i} 3.483(2) (× 2) S(5)–Se(11)^{v} 3.625(2) (× 2)	
B–D	Se(6)–Se(16)^{i} 3.6354(11) (× 2) Se(8)–Se(14)^{i} 3.6383(11) (× 2)	
C–D		S(11)–S(15) 3.552(3)

Symmetry codes:
(i) −x+1, −y+1, −z; (ii) x+1, y, z; (iii) −x+2, −y+1, −z; (iv) −x, −y+2, −z; (v) −x, −y+1, −z; (vi) −x+1, −y, −z

Table 2S. Intra and interstack distances shorter than 3.60 Å (S···S), 3.70 Å (Se···S) or 3.80 Å (Se···Se) (the sum of the van der Waals radii), for the salt $BEST_4[Fe(C_2O_4)_3].C_6H_5COOH\ H_2O$. (**2**).

ET molecules	Interstack	Intrastack
B–B	Se(5)–Se(7)^{iv} 3.6577(13) (× 2)	
C–C	Se(10)–Se(12)^{vi} 3.5891(13) (× 2)	
A–C	Se(1)–Se(11)^{i} 3.5612(13) (× 2) Se(3)–Se(9)^{i} 3.6904(13) (× 2) S(1)–Se(11)^{i} 3.668(2) (× 2)	
A–D	Se(2)–Se(13)^{ii} 3.4883(14) (× 2) Se(4)–Se(15)^{ii} 3.4857(14) (× 2) Se(3)–S(13)^{i} 3.586(3) (× 2) Se(4)–S(15)^{ii} 3.467(2) (× 2) S(2)–Se(13)^{ii} 3.433(2) (× 2) S(4)–Se(14)^{iii} 3.496(2) (× 2)	
B–C	S(8)–Se(10)^{i} 3.464(2) (× 2) S(5)–Se(11)^{v} 3.585(2) (× 2)	
B–D	Se(6)–Se(16)^{i} 3.5959(14) (× 2) Se(8)–Se(14)^{i} 3.6095(14) (× 2) Se(6)–S(16)^{i} 3.683(2) (× 2)	
C–D		S(11)–S(15) 3.488(3)

Symmetry codes:
(i) −x+1, −y+1, −z; (ii) x+1, y, z; (iii) −x+2, −y+1, −z; (iv) −x, −y+2, −z; (v) −x, −y+1, −z; (vi) −x+1, −y, −z

Table 3S. Intra and interstack distances shorter than 3.60 Å (S···S), 3.70 Å (Se···S) or 3.80 Å (Se···Se) (the sum of the van der Waals radii), for the salt $BEST_4[Cr(C_2O_4)_3].1,5H_2O$ (**3**).

ET molecules	Interstack	Intrastack
A−A	Se(1)−Se(2)^{i} 3.5410(16) (× 2) Se(1)−S(2)^{i} 3.686(3) (× 2)	
B−B	Se(3)−Se(4)^{ii} 3.5709(19) (× 2) Se(4)−S(3)^{ii} 3.386(3) (× 2)	
A−B	Se(1)−S(4)^{i} 3.494(3) (× 2) Se(1)−Se(4)^{i} 3.7964(15) (× 2)	
A−B		S(1)−S(3) 3.572(4) S(2)−S(4) 3.597(4)

Symmetry codes:
(i) –x+3/2, −y+1/2, −z+1; (ii) –x+3/2, −y+1/2, −z

Table 4S. Intra and interstack distances shorter than 3.60 Å (S···S), 3.70 Å (Se···S) or 3.80 Å (Se···Se) (the sum of the van der Waals radii), for the salt $BEST_9[Fe(C_2O_4)_3]_2.7H_2O$ (**5**).

ET molecules	Interstack	Intrastack
A−C	Se(1)−S(11)^{i} 3.558(3) (× 2)	
A−B	Se(1)−Se(7)^{i} 3.5605(16) (× 2) Se(3)−Se(5)^{i} 3.5506(17) (× 2) S(3)−Se(5)^{i} 3.681(3) (× 2)	S(4)−S(8)^{iv} 3.530(4) (× 2)
A−D	Se(4)−Se(15) 3.6871(16)	
A−E	Se(2)−S(18)^{ii} 3.368(3) (× 2) Se(2)−Se(18)^{ii} 3.6727(16) (× 2) Se(4)−Se(17)^{iii} 3.4894(16) (× 2) S(4)−Se(17)^{iii} 3.417(3) (× 2)	
C−C	Se(9)−Se(11)^{i} 3.4890(16) (× 2) Se(11)−S(9)^{i} 3.337(3) (× 2)	
B−C		S(5)−S(9) 3.489(4) S(6)−S(10) 3.556(4) S(7)−S(11) 3.570(4)
C−D	Se(10)−Se(16)^{vii} 3.5811(17) (× 2) Se(12)−Se(14)^{vii} 3.791(2) (× 2) Se(12)−S(14)^{vii} 3.690(3) (× 2) S(10)−Se(16)^{vii} 3.608(3) (× 2)	
C−E	Se(12)−S(17)^{iii} 3.372(3) (× 2) Se(12)−Se(17)^{iii} 3.5873(16) (× 2)	
B−B	Se(5)−S(7)^{v} 3.578(3) (× 2)	
B−D	Se(6)−S(13)^{vi} 3.526(3) (× 2) Se(6)−Se(13)^{vi} 3.7058(19) (× 2) Se(8)−S(14)^{vii} 3.443(3) (× 2) Se(8)−Se(14)^{vii} 3.5180(16) (× 2) Se(8)−Se(15)^{vi} 3.7336(17) (× 2) S(6)−Se(16)^{vii} 3.449(3) (× 2) S(8)−Se(15)^{vi} 3.391(3) (× 2)	
D−E		S(13)−S(18)^{ii} 3.530(4) (× 2) S(14)−S(17)^{ii} 3.543(4) (× 2) S(15)−S(17)^{iii} 3.549(4) (× 2) S(16)−S(18)^{iii} 3.558(4) (× 2)

Symmetry codes:
(i) −x+1, −y+1, −z+2; (ii) −x+1, −y, −z+1; (iii) x, y, z+1; (iv) x−1, y, z; (v) −x+2, −y+1, −z+2; (vi) x+1, y, z; (vii) −x+1, −y, −z+2